

STATE-OF-THE-ART REVIEW

Disease Prognosis and Progression in Transthyretin Amyloidosis



JACC: CardioOncology State-of-the-Art Review

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ABSTRACT

Transthyretin amyloidosis (ATTR) is a rapidly evolving disease with a rising prevalence and improving prognosis. With an increasing number of patients diagnosed across a broad spectrum of disease stages, disease prognostication is pivotal to guiding therapeutic decisions. Similarly, with varying phenotypes identified and an expanding range of disease-modifying treatments available, the timely identification of cardiac and extracardiac disease progression remains essential. In this review, the authors highlight the current state of knowledge for disease prognosis and progression in ATTR. They focus on clinical, biomarker, functional, and imaging markers of cardiac disease progression. The authors also highlight the need for further investigation into neurologic and other noncardiac indicators of disease progression. Finally, they discuss current approaches and future directions for disease monitoring in ATTR. (JACC CardioOncol. 2026;8:120-136) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Transthyretin amyloidosis (ATTR) has evolved from a progressive and fatal condition to an increasingly prevalent disease with multiple treatment options. For transthyretin amyloid cardiomyopathy (ATTR-CM), there are now 3 therapies available: tafamidis, acoramidis, and vutrisiran. Similarly, there are 4 drugs in use for transthyretin amyloid polyneuropathy (ATTR-PN): patisiran, vutrisiran, eplontersen, and inotersen, though the latter was removed from the U.S. market. The overarching goal of disease-modifying treatment (DMT) in ATTR is to slow disease progression and improve morbidity and mortality associated with the disease. With increased disease prevalence and a rapidly evolving treatment landscape, accurate prognostication and identification of disease progression in ATTR are pivotal to optimizing the care and guiding the personalized therapeutic path of

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Saro Armenian, MD, Deputy Editor, served as Acting Guest Editor-in-Chief for this paper.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received August 29, 2025; revised manuscript received February 12, 2026, accepted February 13, 2026.

HIGHLIGHTS

- Disease progression assessment in ATTR-CM should be multimodal and globally usable.
- Evidence gaps remain for management of treated patients once progression is identified.
- Multicenter collaborations with patients on DMT needed to address gaps in progression.

patients. Furthermore, it is important for prognostic and progression markers to be straightforward to implement into clinical practice, multimodal, and globally applicable.

CURRENT PROGNOSTIC LANDSCAPE OF ATTR-CM

The prognostic landscape of ATTR-CM is dramatically improving because of earlier diagnosis and the availability of DMT. Key advances that have facilitated earlier diagnosis include increased awareness in the community as well as the development of a noninvasive diagnostic algorithm.¹ This evolution in early diagnosis is exemplified by the improvement in randomized clinical trial placebo group survival, including 57% in ATTR-ACT (Safety and Efficacy of Tafamidis in Patients With Transthyretin Cardiomyopathy; completed in 2018), 74% in ATTRIBUTE-CM (Efficacy and Safety of AG10 in Subjects With Transthyretin Amyloid Cardiomyopathy; completed in 2023), and 82% in HELIOS-B (A Study to Evaluate Vutrisiran in Patients With Transthyretin Amyloidosis With Cardiomyopathy; completed in 2024) at 30 months.²⁻⁴ Importantly, survival data from clinical trials must be interpreted with some caution, as patients may be younger and healthier, with fewer comorbidities, compared with real-world cohorts. Real-world data have also shown improved diagnostic and survival trends, though these too must be interpreted with caution, as most available data have been collected in the pretreatment era. A retrospective observational cohort study from the UK National Amyloidosis Centre (NAC) revealed a dramatic increase in the number of patients diagnosed over a 20-year period (2002-2021). Patients were diagnosed at an earlier disease stage, with an improved median survival from 34.9 months to >60 months despite only 17% of patients in the contemporary cohort being on DMT.⁵ Approved therapies for ATTR-CM have

further elevated the prognostic landscape of the disease, especially when treatment is started at an early disease stage. In THAOS (Transthyretin Amyloidosis Outcomes Survey), survival at 30 months was dramatically better in patients on tafamidis who were enrolled after 2019 (87.3% on treatment vs 77.2% untreated).⁶ Masri et al conducted a multicenter study in the United States of a real-world cohort on treatment with tafamidis, with better survival in this cohort compared with the ATTR-ACT Long Term Extension Study (54.1% vs 44.3%, respectively) at 65 months.^{3,7,8} Beyond early diagnosis, it is possible that the increased use of guideline-directed medical therapy for heart failure (HF), such as mineralocorticoid receptor antagonists and sodium-glucose cotransporter 2 inhibitors, has contributed to more favorable outcomes.⁹⁻¹¹ With improvements in survival over time and the expanding therapeutic armamentarium at our disposal, it is imperative that we turn our attention to identifying prognostic factors that can be used to assist with the initial choice of therapeutic agent and guide patient-centric prognostic discussions.

BASELINE RISK STRATIFICATION OF ATTR-CM

Several individual prognostic factors, broadly related to patient biology and disease severity, have been identified for ATTR-CM. Age at diagnosis, which is affected by the timing of diagnosis and is associated with patients' comorbidities, levels of frailty, and functional status, is an independent predictor of mortality in ATTR-CM.^{8,12,13} The prognosis of ATTR-CM is modified by genotype and degree of cardiac involvement of the specific transthyretin (TTR) variant. The valine-142-isoleucine (p.V142I) variant, present in 3% to 4% of individuals of African and Caribbean descent living in the United States, portends a much worse prognosis compared with other hereditary ATTR genotypes and wild-type ATTR-CM, though whether this is related to the variant itself or to other factors remains to be elucidated.¹⁴⁻¹⁷ Laboratory markers including N-terminal pro-B-type natriuretic peptide (NT-proBNP), troponin T, and estimated glomerular filtration rate have been used to develop 2 key prognostic staging systems in ATTR-CM, the Mayo staging system and the NAC staging system (Table 1).^{18,19} The Columbia risk model

ABBREVIATIONS AND ACRONYMS

- ATTR** = transthyretin amyloidosis
ATTR-CM = transthyretin amyloid cardiomyopathy
ATTR-PN = transthyretin amyloid polyneuropathy
CPET = cardiopulmonary exercise stress testing
DMT = disease-modifying treatment
HF = heart failure
HFH = heart failure hospitalization
KCCQ-OS = Kansas City Cardiomyopathy Questionnaire overall score
LV = left ventricular
NAC = National Amyloidosis Centre
NT-proBNP = N-terminal pro-B-type natriuretic peptide
ODI = oral loop diuretic intensification
PRO = patient-reported outcome
TTR = transthyretin

TABLE 1 Validated Prognostic Staging Systems for Transthyretin Amyloid Cardiomyopathy in the Pretreatment Era

	Mayo (2016) ¹⁹	NAC (2018) ¹⁸	Columbia (2020) ²⁰
Patient population	ATTRwt (N = 360)	ATTRwt (n = 553) ATTRv (n = 316)	ATTRwt (n = 204) ATTRv (n = 105)
Date range for diagnosis	1965-2013	Not included, censor year 2017	2002-2018
Disease-modifying treatment	None	None	None
Parameters	TnT >0.5 ng/mL NT-proBNP >3,000 ng/L	NT-proBNP >3,000 ng/L eGFR <45 mL/min/1.73 m ²	NAC stage (0-2) Daily furosemide equivalent diuretic dose: 0 mg/kg (0), >0-0.5 mg/kg (1), >0.5-1 mg/kg (2), >1 mg/kg (3) NYHA functional class (I-IV)
Stages and median survival	Stage I: 66 mo Stage II: 40 mo Stage III: 20 mo	Original cohort Stage I (45%): 69.2 mo Stage II (38%): 46.7 mo Stage 3 (16%): 24.1 mo	1-3 points: 78 mo 4-6 points: 45.6 mo 7-9 points: 22.8 mo

ATTRv = hereditary transthyretin amyloidosis; ATTRwt = wild-type transthyretin amyloidosis; eGFR = estimated glomerular filtration rate; NAC = National Amyloidosis Centre; NT-proBNP = N-terminal pro-B-type natriuretic peptide; TnT = troponin T.

highlights the importance of baseline clinical factors, including NYHA functional class and diuretic dose, in determining prognosis (Table 1).²⁰ The addition of NYHA functional class and baseline diuretic dose increased the performance of both the Mayo and NAC staging systems for mortality prediction.²⁰ A limitation to the current prognostic staging systems is that patients in the cohorts studied were not on DMT, because of lack of availability. Hence, updated risk models will be important to develop in contemporary patient cohorts.

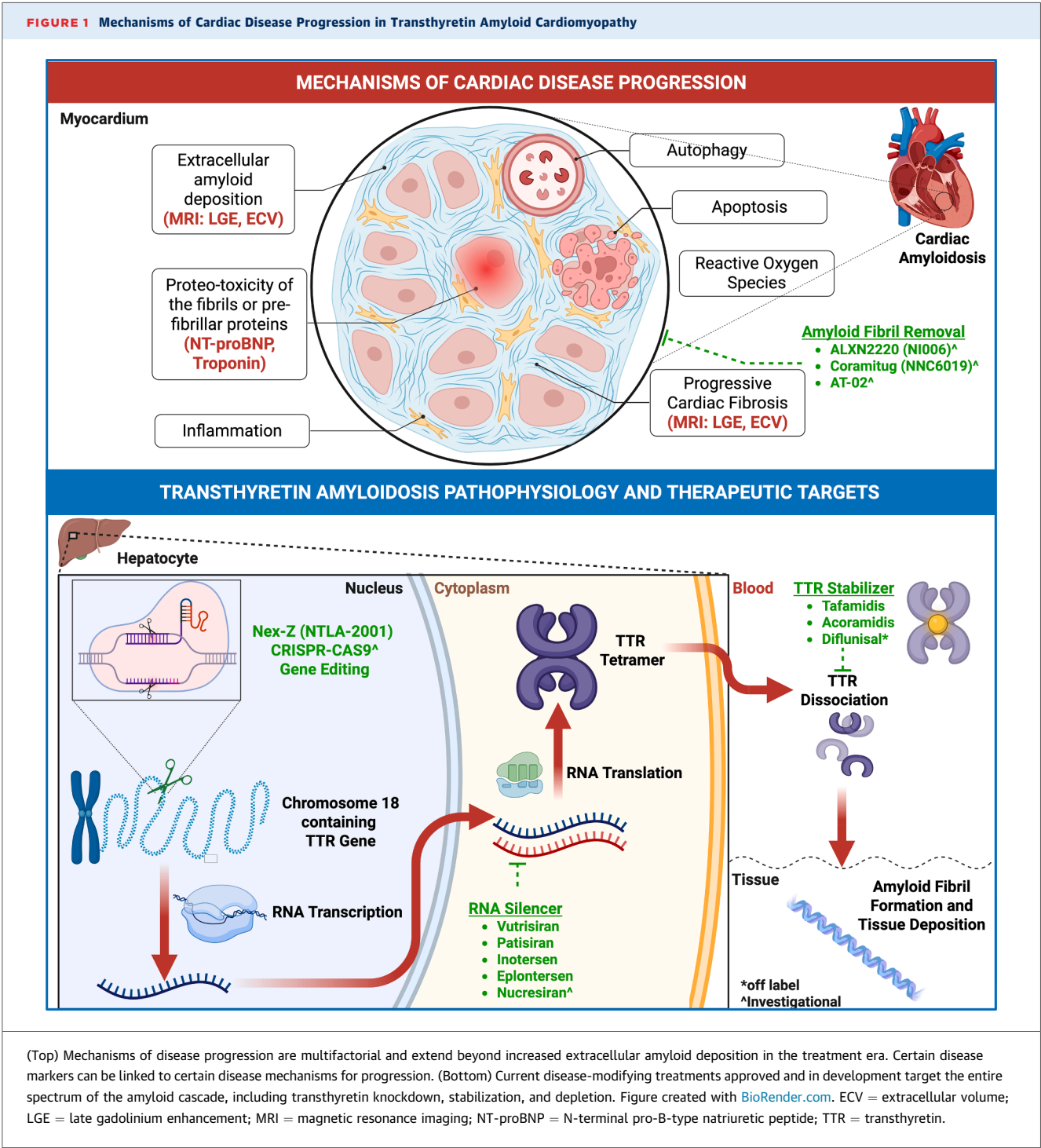
DISEASE PROGRESSION: BACKGROUND, MECHANISMS, AND MANAGEMENT CONSIDERATIONS

Prognostic factors are variables that estimate the likelihood of a certain outcome at time of diagnosis, whereas progression factors are elements measured over time that indicate the worsening of disease. Markers used for disease monitoring have also been used as clinical trial endpoints, broadly defined as prespecified outcome measures to determine whether an intervention has had a meaningful effect. Prognostic and progression factors are multimodal and assessed independently, though disease prognosis and the likelihood of disease progression are inherently linked. In the context of ATTR-CM, disease progression most commonly manifests as worsening HF. In the pre-DMT era, there was continuous amyloid deposition, exerting accelerating deleterious effects on the myocardium. These effects may stem from extracellular amyloid deposition and

proteotoxicity of the fibrils or prefibrillar proteins, leading to inflammation, reactive oxygen species generation, apoptosis, autophagy, and progressive cardiac fibrosis. These processes ultimately interfere with the hemodynamic work load of the heart. However, in the current era, DMT effectively delays amyloid deposition, and disease progression can no longer be attributed to ongoing TTR deposition, though impaired proteostasis, inflammation, autophagy, and fibrosis may still contribute to worsening disease^{14,15,21,22} (Figure 1). Certain markers of disease progression, such as late gadolinium enhancement or extracellular volume on cardiac magnetic resonance imaging, may directly reflect underlying pathogenic mechanisms, whereas others represent the overall worsening of HF. Clinically, it is well established that patients with features of advanced cardiac disease at time of diagnosis deteriorate more rapidly than those with early-stage disease despite treatment with TTR stabilizers or silencers.^{3,16} Currently, there are no data to support switching to a different DMT (eg, stabilizer to silencer or vice versa) or adding an additional DMT in patients experiencing cardiac disease progression, and routine combination of stabilizer and silencer therapy is not recommended.^{17,23}

CARDIAC DISEASE PROGRESSION

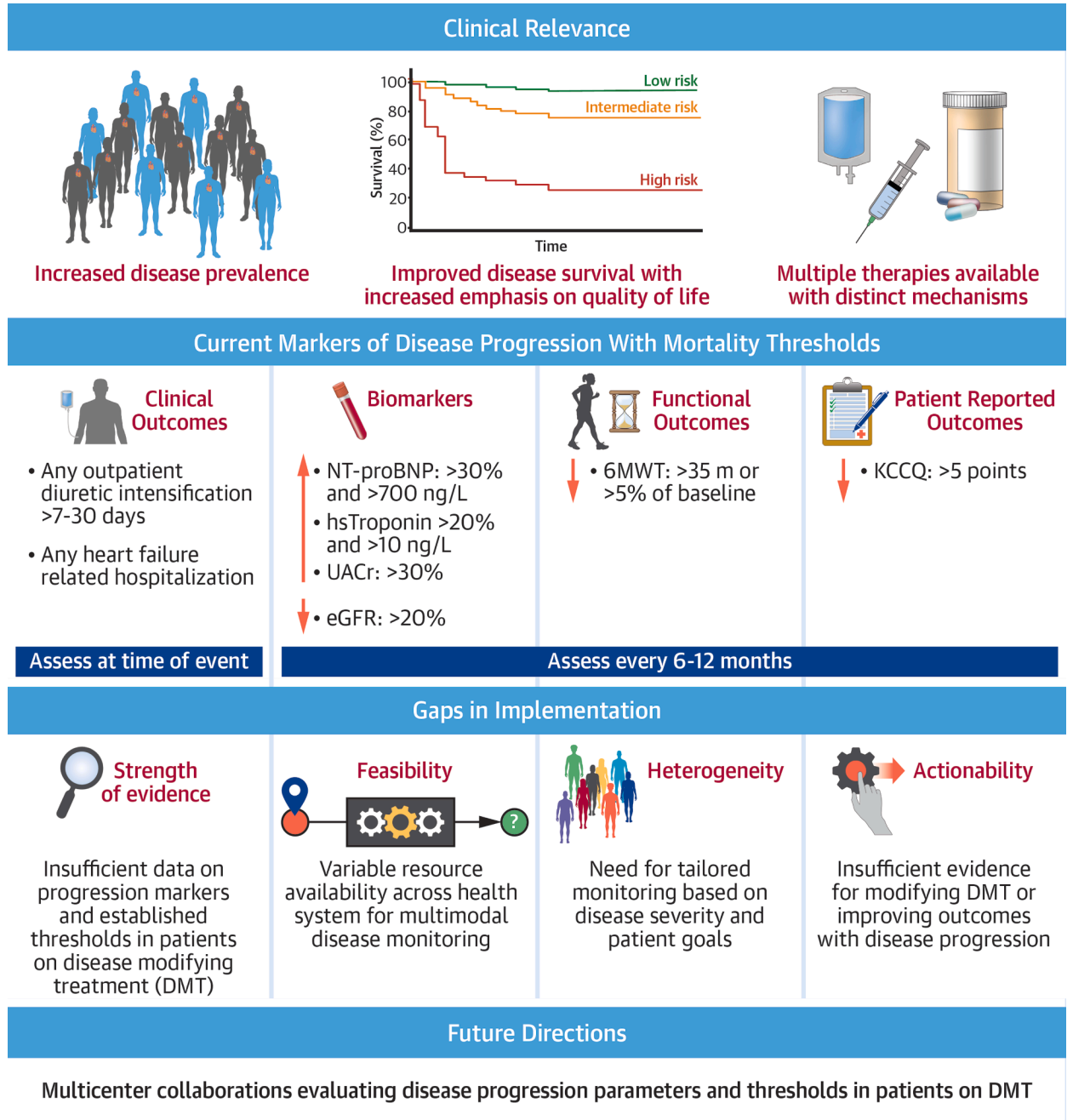
There are domains and individual parameters to consider when evaluating disease progression in ATTR-CM, including clinical markers, biomarkers, functional measures, patient-reported outcomes (PROs), and imaging markers (Central Illustration).



In 2021, an international panel of ATTR-CM experts compiled an expert consensus on disease monitoring and progression incorporating several clinically relevant variables across multiple domains.²⁴ At the time of publication, several of the criteria included had not been validated. Since then, more parameters

have been further investigated and validated. However, a critical limitation of these investigations is that the vast majority of the patients included for study were not on DMT, and thus these data are not truly representative of disease progression and outcomes in patients on DMT (Table 2).

CENTRAL ILLUSTRATION Contemporary Framework, Implementation Gaps, and Future Directions for Evaluation of Disease Progression in Transthyretin Amyloid Cardiomyopathy



Vaishnav J, et al. JACC CardioOncol. 2026;8(2):120-136.

With increasing disease prevalence, improved survival, and a growing number of therapeutic options for transthyretin amyloid cardiomyopathy, rigorous multimodal assessment of disease progression is increasingly necessary. Significant gaps in implementation remain, including limited data in patients receiving disease-modifying therapies, challenges related to feasibility and standardization across diverse health care settings, and the absence of clearly defined management strategies following the identification of disease progression. Multicenter collaborative efforts that incorporate patients on disease-modifying treatment (DMT) will be essential to address these gaps and further advance the field. 6MWT = 6-minute walk test; eGFR = estimated glomerular filtration rate; hsTroponin = high-sensitivity troponin; KCCQ = Kansas City Cardiomyopathy Questionnaire overall score; NT-proBNP = N-terminal pro-B-type natriuretic peptide; UACr = urine albumin-to-creatinine ratio.

TABLE 2 Summary of Studies Evaluating Disease Progression Criteria for Mortality Prediction in Transthyretin Amyloid Cardiomyopathy

	Study	Thresholds for Mortality Prediction	Follow-Up Timeline
Clinical markers			
Outpatient worsening heart failure defined by ODI	- Ioannou et al (2024)²⁵ - n = 515/2,275 on treatment or in trials - Zeldin et al (2024)²⁷ - n = 303 on tafamidis - Bampatsias et al (2024)²⁸ - n = 238 on tafamidis - Fontana et al (2025)²⁹ - n = 366 of 655 on vutrisiran - Nies et al (2025)²⁶ - n = 153 of 182 on tafamidis	Any persistent increase of loop diuretic >7 d or >30 d	12 mo
HF hospitalization	- Zeldin et al (2024)²⁷ - n = 303 on tafamidis	Any time of HF-related hospitalization	Time of hospitalization
Laboratory markers			
NT-proBNP	- Ioannou et al (2024)²⁵ - n = 515/2,275 on treatment or in trials - Bampatsias et al (2024)²⁸ - n = 238 on tafamidis	>30% and >700 ng/L increase	12 mo
Troponin T	- Ioannou et al (2024)²⁵ - n = 515/2,275 on treatment or in trials	>20% and >10 ng/L increase	12 mo
eGFR	- Ioannou et al (2025)⁷⁸ - n = 461/2,001 on treatment or in trials	>20% decline	12 mo
UACR	- Ioannou et al (2024)⁷⁸ - n = 330 patients, treatment NA	>30% increase	12 mo
Patient-reported outcomes			
KCCQ-OS	- Levy et al (2024)⁴⁷ - n = 238 on DMT	>5-point decline	12 mo
Functional measures			
6MWD	- Ioannou et al (2024)³⁸ - n = 1,118 patients, treatment NA	>35-m decline or >5% of baseline	12 mo
CPET	- Patel et al (2025)⁴³ - n = 32 on patisiran, n = 32 matched controls	No established cutoff exists for outcomes Patisiran treatment preserves CPET parameters	12 mo

6MWD = 6-minute walk distance; CPET = cardiopulmonary exercise testing; DMT = disease-modifying treatment; HF = heart failure; KCCQ-OS = Kansas City Cardiomyopathy Questionnaire overall score; NA = not applicable; ODI = oral loop diuretic intensification; UACR = urine albumin/creatinine ratio.

Additional investigation of progression parameters and potential revision of cutoffs are needed in patients on DMT.

CLINICAL MARKERS. Outpatient worsening HF. Although outpatient worsening HF, defined as oral loop diuretic intensification (ODI), was not included in the original expert consensus criteria published in 2021, there are now multiple studies highlighting the significance of this marker. Ioannou et al²⁵ found that 28.2% of UK NAC patients and 44.5% of patients in their external validation cohort experienced ODI by 1 year of follow-up. Baseline risk factors for ODI included older age, the p.V142I variant, and atrial fibrillation. In the UK NAC cohort, ODI was associated with increased risk for mortality (HR: 1.88; 95% CI: 1.62-2.18) in a relationship that was dose dependent, with the highest risk seen in patients who had a >40 mg/d furosemide equivalent increase in dose at 12 months. In a single-center study from Germany including 182 patients with ATTR-CM, of whom 84% were on tafamidis, 25% experienced ODI by 6 months

of follow-up. Patients who had ODI had significantly reduced 2-year survival (72% vs 89%; $P = 0.004$).²⁶ Last, in a cohort followed at Columbia University, the majority of whom were on treatment with tafamidis, ODI (defined as a sustained increase in diuretic dose >1 month) was seen in 47.9% of patients (145 of 303).²⁷ In patients who had both ODI and HF hospitalization (HFH), ODI preceded HFH by more than 5 months and was associated with a higher rate of death (17.7 vs 5.0 per 100 person-years) compared with patients without ODI or HFH over 1.7 years of follow-up. The association was dose dependent, with the highest mortality rates seen in patients who had a ≥ 40 mg/d furosemide equivalent increase compared with a <40 mg/d change (26.64 vs 10.70 per 100 person-years).²⁷ Similar to other studies, ODI remained independently predictive of mortality after adjusting for age and baseline prognostic stage.²⁷ The prognostic significance of ODI for mortality and cardiovascular hospitalization has been validated in an independent cohort on DMT.²⁸ In a subanalysis from

HELIOS-B, outpatient worsening HF was defined as new initiation or sustained increase ≥ 7 days in oral loop diuretic dose.²⁹ Of the 321 patients included, 49.1% had outpatient worsening HF at 36-month follow-up, though vutrisiran use was associated with significantly lower risk for outpatient worsening HF (HR: 0.73; 95% CI: 0.59-0.91). Patients with outpatient worsening HF had a >2 -fold risk for mortality (HR: 2.45; 95% CI: 1.70-3.52). Patients with ODI were more likely to have deteriorations in 6-minute walk distance and Kansas City Cardiomyopathy Questionnaire overall score (KCCQ-OS), as well as increased NT-proBNP, compared with those without diuretic intensification.²⁹ Variability in ODI definitions across existing studies (despite consistent requirements for sustained increases of >7 or >30 days) limits the establishment of a widely accepted validated threshold. Going forward, standardization of duration for ODI will be important when applying this as a progression marker.

HEART FAILURE HOSPITALIZATION. HFH is a well-established adverse event predictive of subsequent mortality in nonamyloid and amyloid-related HF. In an analysis using the Nationwide Readmissions Database, HFH in patients with diagnoses of amyloidosis was associated with a higher rate of inpatient mortality and 30-day readmission rate compared with nonamyloid HF.³⁰ HFH is an important and easily quantifiable measure of disease progression; however, it may be a late manifestation of deterioration and reflect the downstream impact of earlier worsening markers, such as rising NT-proBNP level or ODI. Therefore, it would be ideal to consider this as an outcome measure to be mitigated rather than a progression marker. In the Columbia University cohort, 11.6% of patients with ATTR-CM experienced HFH over 3 years from diagnosis, the majority of which were preceded by ODI.²⁷ HFH was associated with a higher rate of death (29.8 per 100 person-years) compared with ODI alone or no worsening HF events. ATTR-CM clinical trials have focused primarily on cardiovascular-related hospitalization, without distinguishing HF from other cardiovascular events as the cause of hospitalization. Furthermore, the adjudication of HFH may be challenging in the setting of atrial fibrillation-related hospitalization, as the decompensation and increase in NT-proBNP may reflect arrhythmia recurrence rather than worsening HF. Nonetheless, both cardiovascular-related hospitalization and HFH are markers of poor outcomes. In the ATTR-ACT trial, 52.3% of patients on tafamidis had cardiovascular-related hospitalization, of which 43% were HFH.^{3,31} In ATTRibute-CM, 24.7% of

patients on acoramidis had cardiovascular-related hospitalization, with HFH frequency not reported.^{2,32} In both trials, the proportion of cardiovascular-related hospitalization was significantly reduced by DMT (**Table 3**). Going forward, clearer distinctions among cardiovascular-related hospitalization, HFH, and cardiovascular-related events in ATTR-CM trials will help increase precision in evaluation of disease progression.

Summary points.

- Key clinical markers of ATTR-CM disease progression include need for ODI and HFH.
- Sustained ODI of any amount is associated with a 2-fold increased risk in mortality in patients on treatment, though a dose-dependent association has been shown.
- Risk for HFH has decreased over time in patients with ATTR-CM on treatment; however, it is associated with a significantly increased risk for mortality

BIOMARKERS. NT-proBNP. Abnormalities in cardiac biomarkers, including troponin and NT-proBNP, reflect direct toxic effects of amyloid on myocardial cells and myocardial stress from amyloid deposition. Cardiac biomarkers, although not amyloid specific, are the foundation of the major prognostic staging systems in ATTR-CM. Additionally, NT-proBNP has been included as either a composite of the primary endpoint (ATTRibute-CM) or exploratory endpoint (ATTR-ACT, HELIOS-B) in recent ATTR-CM clinical trials (**Table 3**).²⁻⁴ In addition to being a useful prognostic marker, NT-proBNP is a useful marker of progression. An increase in NT-proBNP $>30\%$ and >300 ng/L compared with baseline was included in the expert consensus criteria published in 2021, though without strong evidence to support this threshold. However, there are now more data to guide the use of a specific threshold increase as a progression marker. Ioannou et al²⁵ used the UK NAC cohort as well as an external validation cohort to evaluate the significance of NT-proBNP progression over follow-up. NT-proBNP progression was defined as an absolute increase >700 ng/L and a relative increase $>30\%$. By this definition, approximately one-third of patients in both groups had disease progression at 1 year. NT-proBNP progression was associated with a 1.8-fold higher risk for mortality in both cohorts (NAC: HR: 1.82 [95% CI: 1.57-2.10]; external: HR: 1.75 [95% CI: 1.32-2.33]).²⁵ Fewer than 25% of the patients in this study were on DMT or enrolled in clinical trials. Of

TABLE 3 Summary of Key Outcomes From Contemporary Randomized Clinical Trials in Patients With ATTR-CM

Trial	Medication	Population/ Treatment vs Placebo	Duration	Primary Endpoint	Select Secondary Endpoints
Stabilizers					
ATTR-ACT ³	Tafamidis	ATTR-CM/tafamidis (n = 264) vs placebo (n = 177)	30 mo	Hierarchical (ACM and frequency of CVH); win ratio: 1.7 (95% CI: 1.255-2.289)	- CVH: RR: 0.68 (95% CI: 0.56-0.81) 6MWD: LS mean difference 75.68 m (SE = 9.24 m) - KCCQ-OS: LS mean difference 13.65 points (SE = 2.13 points) - NT-proBNP: Δ at 12 mo = -735.14 ng/L (95% CI: -1,249.16 to -221.13 ng/L); Δ at 30 mo = -2,180.54 ng/L (95% CI: -3,326.14 to -1,034.95 ng/L)
ATTRIBUTE-CM ²	Acoramidis	ATTR-CM/acoramidis (n = 421) vs placebo (n = 211)		Composite hierarchical (all-cause mortality, CVH, NT-proBNP increase, 6MWD); win ratio: 1.8 (95% CI: 1.4-2.2)	- CVH: RR: 0.496 (95% CI: 0.355-0.695) - NT-proBNP geometric mean ratio: 0.529 (95% CI: 0.463-0.604) 6MWD: LS mean difference 39.6 m (95% CI: 21.1-58.2 m) - KCCQ-OS: LS mean difference 9.94 points (95% CI: 5.97-13.91 points)
Silencers					
APOLLO-B ³³	Patisiran	ATTR-CM/patisiran (n = 181) vs placebo (n = 179)	12 mo	Change in 6MWD; mean difference 14.7 m (95% CI: 4.8-24.5 m)	- KCCQ-OS: LS mean difference 3.7 points (95% CI: 0.2-7.2 points) - NT-proBNP: geometric mean ratio: 0.80 (95% CI: 0.73-0.89) - Troponin I: geometric mean ratio: 0.87 (95% CI: 0.80-0.95)
HELIOS-B ⁴	Vutrisiran	ATTR-CM/vutrisiran (n = 326) vs placebo (n = 329)	36 mo (42-mo follow-up)	Composite of all-cause mortality and recurrent cardiovascular events HR: 0.72 (95% CI: 0.56-0.93)	6MWD: LS mean difference 26.5 m (95% CI: 13.4-39.6 m) - KCCQ-OS: LS mean difference 5.8 points (95% CI: 2.4-9.2 points) - NYHA class stability/improvement 68% (vutrisiran) vs 61% (placebo), Δ = 8.7% (95% CI: 1.3%-16.1%)

ACM = all-cause mortality; APOLLO-B = A Study to Evaluate Patisiran in Participants With Transthyretin Amyloidosis With Cardiomyopathy (ATTR Amyloidosis With Cardiomyopathy); ATTR-ACT = Safety and Efficacy of Tafamidis in Patients With Transthyretin Cardiomyopathy; ATTR-CM = transthyretin amyloid cardiomyopathy; ATTRIBUTE-CM = Efficacy and Safety of AG10 in Subjects With Transthyretin Amyloid Cardiomyopathy; CVH = cardiovascular hospitalization; HELIOS-B = A Study to Evaluate Vutrisiran in Patients With Transthyretin Amyloidosis With Cardiomyopathy; LS = least squares; other abbreviations as in Tables 1 and 2.

these, 25.4% had NT-proBNP progression with 3-fold higher risk for mortality. NT-proBNP progression remained an independent predictor of mortality when adjusting for age, NAC stage, and ODI. With the inclusion of both NT-proBNP progression (defined as a >30% increase and a >700 ng/L increase) and ODI, the discriminatory ability of the model improved compared with ODI alone. Notably, fewer than 2% of the cohort were treated with sodium-glucose cotransporter 2 inhibitors, whereas 41% were on mineralocorticoid receptor antagonists and 70% on loop diuretic agents. Therefore, the full impact of sodium-glucose cotransporter 2 inhibitors on NT-proBNP and disease trajectory in ATTR-CM was not fully captured in this study and in the context of disease progression.⁹ The association of NT-proBNP progression >30% and >700 ng/L with mortality and HFH has since been independently validated in a cohort of patients on tafamidis treated at Columbia University.²⁸ In that analysis, the prognostic value of change in NT-proBNP was evaluated. Change in NT-proBNP was independently

associated with mortality after adjustment for baseline NT-proBNP, age, estimated glomerular filtration rate, and ODI. However, this association was no longer significant after additional adjustment for follow-up NT-proBNP. In contrast, follow-up NT-proBNP remained independently associated with mortality when adjusted for age, estimated glomerular filtration rate, and ODI. These findings suggest that although change in NT-proBNP is prognostic, its predictive value is largely captured by the follow-up NT-proBNP level. Nevertheless, these findings underscore the importance of longitudinal monitoring of NT-proBNP in ATTR-CM.

Troponin. The use of troponin as a marker of disease progression is cumbersome because of the variety of troponin assays available in clinical practice. Some laboratories use troponin T, whereas others use troponin I, and more recently there has been a shift toward using high-sensitivity troponin assays. Change in troponin I was included as an exploratory endpoint in APOLLO-B (A Study to Evaluate Patisiran in Participants With Transthyretin Amyloidosis With

Cardiomyopathy [ATTR Amyloidosis With Cardiomyopathy]), with a smaller median change from baseline to 1 year in the patisiran compared with placebo group (3.8 ng/L vs 14.5 ng/L).³³ Similarly, in an exploratory endpoint of HELIOS-B, there was a smaller change in troponin I in the vutrisiran group compared with placebo (geometric fold-change ratio: 0.68; 95% CI: 0.62-0.75).⁴ Increase in troponin was included in the 2021 expert consensus guidelines as a marker of disease progression, though without data to guide this recommendation. Since then, Ioannou et al²⁵ have investigated troponin T as a marker of disease progression, with the optimal threshold established to be an increase of 10 ng/L (sensitivity, 65.7%; specificity, 63.7%) and a percentage change of 20.3% (sensitivity, 63.2%; specificity, 59.1%). At 1 year, 231 patients (50.3%) experienced troponin T progression, which was associated with a 1.8-fold higher risk for mortality. As troponin elevation reflects both amyloid toxicity and myocardial stress in ATTR-CM, future studies are needed to determine whether changes in troponin track amyloid deposition, myocardial stress, or both. Compared with NT-proBNP, the widespread implementation of troponin as a marker of disease progression may be limited by the multiple troponin assays used worldwide.

Serum Transthyretin. Dissociation of the TTR tetramer into amyloidogenic monomers is the rate-limiting step in the amyloidogenesis cascade. Serum TTR, also known as prealbumin, has been studied as an indirect measure of TTR stabilization and a prognostic marker.³⁴ A recent subanalysis from ATTRIBUTE-CM highlighted the prognostic significance of this marker, with a low baseline serum TTR (<20 mg/dL) associated with a significantly higher risk for mortality at 30 months.³⁵ Furthermore, an early (within 28 days) increase in serum TTR was incrementally associated with reduced risk for mortality.³⁵ Specifically, there was a 25% relative reduction in risk for mortality for each 5 mg/dL increase in early serum TTR after treatment initiation. The analysis also showed that treatment-related increases in serum TTR mediated the effect of acoramidis treatment on decreasing mortality. These data suggest that serum TTR levels may be a useful biological surrogate for in vivo TTR stabilization, particularly early after treatment initiation. However, because of certain limitations with serum TTR, including the lack of specificity as well as potential to be influenced by various factors such as age, sex, nutritional status, body composition, and inflammation, its use as a disease progression marker may be limited when applied to an individual patient.

Additionally, serum TTR levels are reduced with gene-silencer therapy, which constrains their usefulness as a disease marker both in this context and among patients receiving combination therapy.

Summary points.

- NT-proBNP is key biomarker used for disease prognostication and evaluation of progression in patients with ATTR-CM.
- An increase in NT-proBNP >30% and >700 ng/L has been externally validated as a marker of disease progression associated with a 2-fold increase in mortality.
- An increase in troponin T of >20% and >10 ng/L is associated with an approximately 2-fold increase in mortality, though widespread use of troponin as a disease marker may be limited because of multiple assays used worldwide.
- Serum TTR levels increase with stabilizer therapy, and these increases are associated on a population level with improved survival, but application to an individual patient may be affected by other confounders that affect serum TTR levels.

FUNCTIONAL MEASURES. 6-minute walk test. The 6-minute walk test provides an assessment of cardiopulmonary response to exercise. The attraction of this test lies in its reproducibility, validity, low cost, and ease of administration.³⁶ Six-minute walk test data from 2,141 patients with ATTR-CM showed a 2.2-fold higher risk for mortality in those who were unable to complete 350 m at baseline, with similar risk for all NAC stages and genotypes, highlighting use as a prognostic marker of disease.^{37,38} A 30- to 40-m decline was included as a threshold value for disease progression in the 2021 expert consensus guidelines. Since then, data from the NAC have shown a threshold of 35 m or 5% decrease from baseline distance at 1 year to be significant, as it is associated with an almost 2-fold increased risk for mortality.³⁸ Despite its simplicity, the 6-minute walk test does have limitations, most particularly in that it does not provide information on the mechanism of disease limitation (ie, cardiac, pulmonary, neurologic, or orthopedic), which may be delineated with more objective testing. Additional limitations of the 6-minute walk test more specific to the ATTR-CM patient population may also include comorbidities such as spinal stenosis and gait instability.

Cardiopulmonary exercise stress testing. Cardiopulmonary exercise testing (CPET) is the gold-standard diagnostic modality to evaluate a patient's functional capacity. Peak oxygen uptake is a well-known predictor of mortality in patients without amyloidosis awaiting heart transplantation, with a survival

rate of only 70% in those with peak oxygen uptake ≤ 14 mL/kg/min.³⁹ In patients with stable HF with peak oxygen uptake > 18 mL/kg/min, a minute ventilation/carbon dioxide production slope of > 34 was associated with a 4-fold increased risk for mortality at 3 years.⁴⁰ As a prognostic marker, percentage predicted peak oxygen uptake is significantly reduced and correlates closely with the degree of cardiac dysfunction, on the basis of biomarker abnormalities and cardiac stage, in both light chain amyloidosis and ATTR-CM.⁴¹ Data from the NAC evaluated CPET in patients with ATTR-CM, mixed ATTR, and ATTR-PN (most of whom were not on DMT), demonstrating that impairment in functional capacity is highly prevalent.⁴² Mean peak oxygen uptake, percentage predicted oxygen uptake, and peak work load were significantly lower in ATTR-CM and mixed ATTR than in ATTR-PN. Mean minute ventilation/carbon dioxide production slope was significantly higher in patients with ATTR-CM (38.1) compared with those with mixed ATTR (34.0) and ATTR-PN (28.7). Furthermore, percentage predicted peak oxygen uptake and minute ventilation/carbon dioxide production slope were more severely impaired with increased amyloid burden as measured by extracellular volume on cardiac magnetic resonance imaging. Peak oxygen uptake and peak systolic blood pressure were independent predictors of mortality.⁴² CPET parameters were not incorporated into expert consensus criteria on disease prognosis or progression in ATTR-CM, though this has more recently been a topic of interest. A small prospective study evaluating change in CPET parameters over 1 year showed a greater decline in percentage predicted peak oxygen uptake (-6.2% vs $+0.3\%$) and ventilatory efficiency (minute ventilation/carbon dioxide production; $+0.9$ vs -2.7) in the untreated group compared with those on patisiran. Other CPET parameters were similar between both groups.⁴³ Similar ATTR-CM-specific comorbid limitations may exist for feasibility of CPET as for the 6-minute walk test. Additional studies investigating serial changes in CPET in ATTR-CM patients on DMT are needed to power a future clinical trial.

Summary points.

- There are limited observational data on 6-minute walk distance as a marker of disease progression, though a decrease of 35 m or 5% from baseline in untreated patients has been associated with a near 2-fold increased risk for mortality.
- CPET is an objective and reproducible tool for the evaluation of disease prognosis and progression in ATTR-CM, but it is underused, and further study is needed to define clinically relevant thresholds.

PATIENT-REPORTED OUTCOMES. PROs are measures directly reported by patients; they focus on patients' health, quality of life, or functional status regarding health care or treatment.⁴⁴ In recent ATTR-CM clinical trials, PROs have been used as secondary outcome measures, establishing that DMT mitigates decline in key PROs. In the ATTR-ACT trial, PROs collected included the KCCQ-OS, the EQ-5D-3L, the EuroQol Visual Analogue Scale, and the patient global assessment of overall health score.³ The placebo group experienced a decline in KCCQ-OS over 30-month follow-up, with an average decline of 14.6 (21.4) points, less pronounced in wild-type ATTR compared with hereditary ATTR.^{3,45,46} Treatment with tafamidis reduced the decline in KCCQ-OS compared with placebo (mean difference 13.65), with a difference seen as early as 6 months into follow-up (Table 3). Furthermore, treatment with tafamidis led to a higher proportion of patients either having an improvement or no change in KCCQ-OS over follow-up compared with placebo (41.8% vs 21.4%).⁴⁵ Similarly, in ATTRIBUTE-CM and HELIOS-B, patients in the treatment arm had less of a decrement in KCCQ-OS over follow-up compared with placebo (least squares mean differences of 9.94 and 5.8, respectively).^{2,4} Similar findings were noted in the ATTR-ACT trial for the EQ-5D-3L, EuroQol Visual Analogue Scale, and patient global assessment scores.⁴⁵

There are limited real-world data on PROs in the context of disease progression. Levy et al⁴⁷ found that 27% of 201 patients on tafamidis had > 5 -point decreases in KCCQ-OS within 1 year, which were associated with significantly higher mortality or cardiovascular hospitalization risk compared with patients who had stable KCCQ-OS. Worsening of KCCQ-OS is independently associated with mortality when adjusted for genotype, NAC stage, NT-proBNP increase, and increased diuretic dose.⁴⁷ PROs were not included in the 2021 expert consensus guidelines for the evaluation of disease progression, though they serve as a useful and important marker of progression. PROs, as opposed to other disease progression markers discussed, capture quality of life, which is increasingly important in the ATTR-CM patient population who are diagnosed earlier, are less subject to highly morbid disease outcomes, and are living longer with the disease. Further studies validating these findings and examining the utility of other PROs for disease monitoring are needed.

Summary points.

- PROs have been incorporated as clinical trial endpoints, though there are limited real-world

observational data as markers of disease progression in ATTR-CM.

- In randomized clinical trials, treatment with stabilizers or silencers attenuate the decrement in KCCQ-OS.
- Given the ability to capture quality of life and symptom burden, more data on PROs as markers of disease progression from observational cohorts are needed as patients are living longer with ATTR-CM.

ELECTROCARDIOGRAPHIC AND IMAGING MARKERS.

Electrocardiography. Electrocardiography has a fundamental role in suspicion for ATTR-CM, though it also lends valuable information on disease prognosis. Studies to date have been limited by small sample sizes, though they have consistently shown that low QRS voltage is negatively prognostic in cardiac amyloidosis.⁴⁸⁻⁵⁰ In a multicenter analysis, Cipriani et al⁴⁹ found that patients with low QRS voltages had worse survival over follow-up (80% vs 95% at 40 months; $P = 0.009$). After adjustment for markers of disease severity, low QRS voltage remained predictive of mortality. Additionally wide QRS duration at baseline has been associated with adverse outcomes, with one analysis showing a significant difference in survival according to QRS interval >120 ms or <120 ms.⁵¹ Atrial fibrillation, although exceedingly common in ATTR-CM, has not been shown to be independently prognostic in single-center and post hoc analyses to date.⁵²⁻⁵⁴ Robust data on associations of electrocardiographic parameters with disease progression and clinical outcomes are lacking. Similarly, additional information is needed on the burden of atrial fibrillation in patients on DMT. Future, larger collaborations will be helpful to inform use of electrocardiographic changes as a disease progression monitor as well as attenuation of arrhythmias on disease outcomes. Furthermore, there is likely a significant role in applying artificial intelligence to electrocardiographic interpretation for screening, prognostication, and disease monitoring of ATTR-CM.

Echocardiography. Transthoracic echocardiography is the most frequently performed diagnostic imaging modality for ATTR-CM. In a prospective longitudinal study from the NAC, prognostic and progression markers on transthoracic echocardiography were assessed in 877 untreated patients with ATTR-CM at baseline, 12 months, and at 24 months.⁵⁵ Prognostic echocardiographic markers, predictive of mortality, included mitral regurgitation, severe

tricuspid regurgitation, severe aortic stenosis, and stroke volume index.⁵⁵ Most transthoracic echocardiographic parameters pertaining to left ventricular (LV) concentric remodeling, atrial enlargement, systolic function, and diastolic function significantly worsened in the cohort. In multivariable models, mitral regurgitation progression and stroke volume index change had independent prognostic significance. Myocardial deformation measures, including global longitudinal strain as well as myocardial work, have been shown to be prognostic in patients with ATTR-CM.⁵⁶

Randomized clinical trials have evaluated change in echocardiographic parameters over time. In a subanalysis from the ATTR-ACT trial, treatment with tafamidis led to a smaller decline in echocardiographic parameters, including LV ejection fraction, LV stroke volume, LV global longitudinal strain, and septal E/e' ratio. Single-center studies have similarly shown attenuated decline in cardiac function with tafamidis treatment.⁵⁷⁻⁵⁹ Changes in echocardiographic parameters were investigated in a subanalysis from HELIOS-B, showing favorable effects on attenuation of changes in LV wall thickness, LV mass index, LV ejection fraction, global longitudinal strain, and LV stroke volume. Although these findings were overall positive and statistically significant compared with placebo, absolute differences between groups were relatively modest.⁶⁰ Regression in imaging markers with antifibril therapy is an area of ongoing investigation and has been incorporated into the study protocols. In a recent phase 2 trial of coramitug vs placebo, benefits were seen with high-dose coramitug (60 mg/kg) compared with placebo on exploratory echocardiographic parameters, including LV stroke volume index, mitral A-wave velocity, left atrial end-systolic volume, and right ventricular systolic pressure; their significance will be further evaluated in the ongoing phase 3 trial.⁶¹ Although changes in echocardiographic parameters including LV wall thickness, diastolic dysfunction, and systolic dysfunction were incorporated in the 2021 expert consensus guidelines for disease monitoring, there are no validated echocardiographic criteria for disease progression on treatment for ATTR-CM. The use of echocardiographic measures as metrics for disease progression may be limited by high interobserver variability in measurements in the setting of relatively small changes in these parameters noted from studies to date. Furthermore, transthoracic echocardiographic parameters may be influenced by temporal factors such as loading conditions of the LV. Until these limitations are overcome and more high-quality data are generated on the progression of

echocardiographic parameters on treatment, there is limited utility in use as a disease progression marker. **CMR.** Cardiac magnetic resonance imaging is a well-established diagnostic aid for cardiac amyloidosis. Given its ability to provide robust tissue characterization and visualization of amyloid infiltration with late gadolinium enhancement, the use of cardiac magnetic resonance imaging has also been investigated for prognostication in both light chain amyloidosis and ATTR-CM. In a study including 263 patients diagnosed with ATTR-CM by nuclear scintigraphy, there was a correlation between extracellular volume and uptake on ^{99m}Tc 3,3-diphosphono-1,2-propanodicarboxylic acid imaging ($r = 0.533$) and varied across different variant subtypes.⁶² During follow-up, extracellular volume was independently associated with mortality when adjusted for age, NT-proBNP, ejection fraction, E/e' ratio, and LV mass index (HR: 1.164; 95% CI: 1.066-1.271). Although cardiac magnetic resonance imaging has not been routinely performed as part of recent clinical trials, ongoing trials investigating the safety and efficacy of antifibril therapies are using cardiac magnetic resonance imaging. There are limited data on cardiac magnetic resonance imaging parameters as disease progression markers in patients on therapy. In a small study of 36 patients on tafamidis and 15 untreated patients, extracellular volume or T1 mapping did not change in either group at 12 months. However, patients in the tafamidis group had significantly reduced LV mass, whereas the group on no treatment had worse LV global longitudinal strain over time.⁶³ Fontana et al⁶⁴ investigated longitudinal change in extracellular volume in 32 patients, of whom 50% were on treatment with patisiran. At 12-month follow-up, extracellular volume was on average 6.2% lower in patients on treatment after adjusting for baseline extracellular volume. Widespread implementation of cardiac magnetic resonance imaging for monitoring ATTR-CM disease progression may be limited by patient tolerance, accessibility, and variation in imaging protocols and expertise across centers.

Summary points.

- In untreated patients, echocardiographic parameters associated with worse prognosis include mitral or tricuspid regurgitation, severe aortic stenosis, and reduced stroke volume index.
- Randomized clinical trials with stabilizers and silencers have shown marginal change in echocardiographic or cardiac magnetic resonance imaging parameters over time in treated patients, though additional data are expected from ongoing studies using antifibril therapy in ATTR-CM.

- The role of electrocardiographic and imaging markers of disease progression in addition to other prognostic variables is unclear.

PERIPHERAL NERVOUS SYSTEM DISEASE PROGNOSIS AND PROGRESSION

Peripheral nervous system involvement has traditionally been the defining feature of the “neurologic phenotype” of hereditary ATTR. The presence or absence of neuropathy has guided the selection of DMT in ATTR until recently.⁶⁵ However, with the expanding landscape of emerging treatments for both ATTR-PN and ATTR-CM, there is a growing need for robust data to inform therapeutic decision-making and optimize individualized treatment strategies in the future.

ASSESSMENTS FOR NEUROLOGIC DISEASE MONITORING. Results from multiple clinical trials have demonstrated the effectiveness of DMT in halting the progression of polyneuropathy in patients with hereditary ATTR-PN (Table 4).⁶⁶⁻⁶⁹ There are several tools to quantify neurologic impairment in ATTR-PN, including clinician assessments such as the Neuropathy Impairment Score-Lower Limbs and the modified Neuropathy Impairment Score +7, PRO questionnaires such as the Norfolk Quality of Life-Diabetic Neuropathy questionnaire and the Composite Autonomic Symptom Score-31, and the Neuropathy Symptoms and Change score, which combines objective and subjective measures.

In 2011, tafamidis became the first DMT to be approved for ATTR-PN in Europe following a phase 3 trial in p.V50M patients. Although tafamidis delayed the rate of neurologic decline, mean neuropathy scores still worsened over time, and the therapeutic benefits were most evident in early-stage disease.^{70,71} Notably, for ATTR-PN, tafamidis has not been evaluated in larger randomized trials, nor is it approved for this indication in the United States. TTR gene silencers have shown efficacy in randomized controlled trials and are the standard of care for ATTR-PN in the United States.⁶⁶⁻⁶⁹ Findings from ATTR-PN studies to date consistently demonstrate meaningful benefits of RNA silencer therapy in neurologic function, quality of life, and nutritional status across multiple trials (Table 4).

Despite the advances in therapeutic modalities, there are no guidelines or established protocols for monitoring progression in patients with ATTR-PN with or without ATTR-CM.⁷² Recent findings from NEURO-TTTransform (A Study to Evaluate the Efficacy and Safety of Eplontersen [Formerly Known as

TABLE 4 Summary of Key Outcomes in Contemporary Clinical Trials in Patients With ATTR-PN

Trial	Medication	Population/Treatment vs Placebo	Duration	Primary Endpoint	Select Secondary Endpoints
Fx-1006A ⁷¹	Tafamidis	ATTR-PN/tafamidis (n = 65) vs placebo (n = 60)	18 mo	mNIS and mBMI	Significant delay in neurologic progression; better preservation of nerve function
NEURO-TTR ⁶⁷	Inotersen	ATTR-PN/inotersen (n = 113) vs placebo (n = 60)	15 mo	mNIS+7 Δ −19.7 (95% CI: −26.4 to −13.0; P < 0.001)	Norfolk QOL-DN Δ −11.7 (95% CI: −18.3 to −5.1)
APOLLO ⁶⁸	Patisiran	ATTR-PN/patisiran (n = 148) vs placebo (n = 77)	18 mo	mNIS+7 Δ −34.0 (95% CI: −39.9 to −28.1; P < 0.001)	Norfolk QOL-DN −6.7 vs +14.4; Δ −21.1 (95% CI: −27.2 to −15.0) Reduced NFL significantly
HELIOS ⁶⁶	Vutrisiran	ATTR-PN/vutrisiran (n = 122) vs historical placebo (n = 77)	18 mo	mNIS+7 Δ −28.5 (95% CI: −34.0 to −23.1; P < 0.001)	Norfolk QOL-DN Δ −21.0 (95% CI: −27.1 to −14.9) mBMI preserved
NEURO-TTRtransform ⁶⁹	Eplontersen	ATTR-PN/eplontersen (n = 168) vs historical placebo (n = 65)	66 wk	mNIS+7 +0.3 vs +25.1; Δ −24.8 (95% CI: −31.0 to −18.6; P < 0.001)	Norfolk QOL-DN −5.5 vs +14.2; Δ −19.7 (95% CI: −25.6 to −13.8) mBMI stabilized

ATTR-PN = transthyretin amyloid polyneuropathy; mBMI = modified body mass index; mNIS = modified Neuropathy Impairment Score; NFL = neurofilament light chain; Norfolk QOL-DN = Norfolk Quality of Life-Diabetic Neuropathy.

ION-682884, IONIS-TTR-LRx and AKCEA-TTR-LRx] in Participants With Hereditary Transthyretin-Mediated Amyloid Polyneuropathy) have proposed thresholds for clinically meaningful change in ATTR-PN outcomes.⁷³ For the modified Neuropathy Impairment Score +7, improvement was defined as a decrease of ≥ 4.0 points and worsening as an increase of ≥ 1.8 points at 85 weeks. For the Norfolk Quality of Life-Diabetic Neuropathy questionnaire, meaningful improvement ranged from −12.8 to −1.4 points and deterioration from +5.9 to +14.7 points at 37 and 85 weeks. Changes in the Neuropathy Symptom and Change score of −2.4 to −1.3 points indicated improvement at weeks 37 and 85, while changes of +0.6 to +5.8 points signified worsening at weeks 35, 66, and 85. Although the modified Neuropathy Impairment Score +7 remains the most comprehensive tool, it is resource intensive, making it less practical for widespread use outside of research centers. More feasible alternatives, such as the Neuropathy Impairment Score-Lower Limbs for clinical assessment, along with focused PROs such as the Norfolk Quality of Life-Diabetic Neuropathy questionnaire and Composite Autonomic Symptom Score-31, may be more practical in everyday clinical settings. A multimodal approach combining physician-administered neurologic assessments and PROs to capture both objective deficits and subjective symptoms may enhance disease monitoring.⁷² Integrating these parameters into routine care at 6- to 12-month intervals may support earlier recognition of disease progression.

BIOMARKERS FOR NEUROLOGIC DISEASE MONITORING.

In contrast to ATTR-CM, there is a lack of

standardized biomarkers for ATTR-PN. Neurofilament light chain may serve as a promising biomarker of axonal injury and treatment response, although it is not clear if it enables earlier detection of subclinical progression.⁷⁴ Although neurofilament light chain is not an amyloid-specific marker of peripheral neuropathy, elevated serum levels reflect ongoing axonal degeneration and have been shown to correlate with disease severity and progression in ATTR-PN.⁷⁴⁻⁷⁶ Optimal cutoff values for serum neurofilament light chain levels in ATTR-PN are still under investigation, and the application as prognostic or treatment sensitive biomarker in clinical practice is not clear yet. Further studies are needed to develop novel biomarkers for ATTR-PN as well as establish optimal cutoffs, timing, and standardized reference values.

SUMMARY POINTS.

- There are no validated criteria or previously published expert consensus guidelines on neurologic disease progression in patients with ATTR-PN.
- The modified Neuropathy Impairment Score +7 is a comprehensive clinical evaluation that can be used to monitor disease progression in patients with ATTR-PN.

OTHER ORGAN SYSTEMS

With ATTR being a systemic disease, in theory any organ system may be involved and warrant ongoing monitoring. The mechanisms of chronic kidney disease in ATTR-CM include sequelae of cardiorenal syndrome and other potential comorbidities and, to a lesser degree, direct renal amyloid deposition.⁷⁷ Renal function as measured by estimated glomerular

filtration rate is well established as a baseline prognostic factor in ATTR-CM. Additionally, the presence of albuminuria at diagnosis (defined as urine albumin-to-creatinine ratio > 30 mg/g) is associated with more severe disease, and a greater degree of albuminuria at diagnosis is independently associated with worse survival.⁷⁸ In terms of renal disease progression, an increase in urine albumin-to-creatinine ratio of $\geq 30\%$ over a 1-year period was shown to predict mortality.⁷⁸ In a large single-center retrospective study of patients with ATTR-CM, a decline in estimated glomerular filtration rate of $>20\%$ over a 1-year period was associated with increased risk for mortality, with consistent findings across ATTR-CM genotypes and NAC stages and among those on DMT.⁷⁹ A post hoc analysis of ATTR-ACT showed that tafamidis stabilized the decline in renal function over time compared with placebo.⁸⁰

Among patients with ATTR-CM, hyperbilirubinemia, elevated alkaline phosphatase, and elevated gamma-glutamyl transferase have been associated with increased risk for mortality, likely reflecting worsening right HF and tricuspid regurgitation.^{81,82} Other gastrointestinal manifestations of ATTR amyloidosis have mainly been associated with hereditary ATTR and are related predominantly to autonomic neuropathy and dysmotility.⁸³ Modified body mass index was developed as a measure to better capture malnutrition in patients with ATTR-PN in the setting of edematous states and is associated with death in this population, though has not been shown to be an independent predictor of mortality in ATTR-CM.^{84,85}

Musculoskeletal manifestations, including carpal tunnel syndrome and lumbar spinal stenosis, are well established in ATTR, though they frequently precede cardiac involvement by several years; therefore, their utility for assessing disease progression is limited. Impairment in other organ systems, including pulmonary and hematologic as well as inflammatory markers, has been studied largely in the context of ATTR-CM; consequently, distinguishing abnormalities attributable to TTR deposition from sequelae of HF can be challenging. Certainly, more research is needed to validate noncardiac markers of disease progression in ATTR as well as disentangle specific mechanisms of impairment.

IMPLEMENTATION AND FUTURE DIRECTIONS

Numerous progression factors have been identified that provide insights into the systemic nature of ATTR. Several of these markers of disease progression have been studied in clinical trials as well as in

observational cohorts of affected patients, and some have been subjected to validation in independent cohorts. As some of our current data indicate, the presence of multiple markers of disease progression across different domains likely associates with worse overall prognosis, underscoring the importance of multimodal assessment. Although this represents progress in the field, there are implementation gaps that remain (**Central Illustration**). Specifically, the need for more robust data in patients receiving DMT as well as ensuring feasibility and standardization globally, across diverse health care settings. Patients and caregivers often express a desire to understand the stability of their disease in the context of both quality of life and survival. Disease monitoring must therefore include direct, ongoing dialogue with patients, leveraging advocacy and support group organizations to ensure their input drive care. From a provider perspective, there is a lack of clearly defined, actionable strategies once disease progression occurs. Future multicenter, international collaboration will be important to define a parsimonious set of progression factors, as well as evidence-based management strategies, which can inform the care of affected patients in order to achieve our collective goal of enhancing functional capacity, improving quality of life, and reducing morbidity and mortality.

ACKNOWLEDGMENTS The authors thank the members of the Amyloidosis Research Consortium Prognostic Factors Working Group.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Vaishnav is a consultant, speaker, and advisory board member for Pfizer and Alnylam Pharmaceuticals. Dr Boursiquot has received support from National Institutes of Health postdoctoral training grant T32HL007854. Dr Chakraborty is a consultant and advisory board member for Janssen, Sanofi, Adaptive Biotech, and Alexion (AstraZeneca); and has received research funding from Genentech and AbbVie. Dr Wechalekar is an advisory board member for and has received honoraria from Janssen, GlaxoSmithKline, Prothena, Alexion/AstraZeneca, and Attralus. Dr Garcia Pavia has received speaker fees from Alnylam Pharmaceuticals, AstraZeneca, Bayer, BridgeBio, Intellia, Ionis Pharmaceuticals, Novo Nordisk, and Pfizer; has received consulting fees from Alexion, Alnylam Pharmaceuticals, AstraZeneca, Attralus, Bayer, BridgeBio, Intellia, Ionis Pharmaceuticals, Life Molecular Imaging, Pfizer, Neuroimmune, and Novo Nordisk; and has received research and educational support to his institution from Alnylam Pharmaceuticals, AstraZeneca, Bayer, BridgeBio, Intellia, Novo Nordisk, and Pfizer. Dr Witteles is an adviser for Pfizer, Alnylam, BridgeBio, Alexion, and AstraZeneca. Dr Hanna is an advisory board member for Pfizer, Alnylam Pharmaceuticals, BridgeBio, Ionis, Attralus, Alexion Pharmaceuticals, and Novo Nordisk. Dr Karam has received grant support from AstraZeneca, Argenx, Ionis, and Sanofi/Genzyme; has received consulting fees from Alnylam Pharmaceuticals, AstraZeneca, Ionis, Sanofi, Takeda, Argenx, and Alexion; and is an advisory board member for Alexion, Alnylam Pharmaceuticals, Alpine Pharmaceuticals, Amgen,

Amicus Pharmaceuticals, Annexon, argenx, AstraZeneca, Biogen, Corino Therapeutics, CSL Behring, Genentech, Ionis, Johnson & Johnson, NeuroDerm, Novo Nordisk, Octapharma USA, Sanofi, Pfizer, UCB, Takeda, and Zai Lab. Dr Maurer has received grant support from National Institutes of Health grants R01HL177670 and R01AG081582; has received grants and personal fees from Alnylam Pharmaceuticals, Pfizer, Eidos, Prothena, and Ionis; and has received personal fees from AstraZeneca, Akcea, Intellia, and Novo Nordisk. Dr Griffin is an advisory board member for BridgeBio, AstraZeneca, and Pfizer; has received grants from Pfizer and BridgeBio; and is a Speakers Bureau member for BridgeBio. All other authors have reported that

they have no relationships relevant to the contents of this paper to disclose.

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REFERENCES

- Gillmore JD, Maurer MS, Falk RH, et al. Non-biopsy diagnosis of cardiac transthyretin amyloidosis. *Circulation*. 2016;133:2404-2412.
- Gillmore JD, Judge DP, Cappelli F, et al. Efficacy and safety of acoramidis in transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2024;390:132-142.
- Maurer MS, Schwartz JH, Gundapaneni B, et al. Tafamidis treatment for patients with transthyretin amyloid cardiomyopathy. *N Engl J Med*. 2018;379:1007-1016.
- Fontana M, Berk JL, Gillmore JD, et al. Vutrisiran in patients with transthyretin amyloidosis with cardiomyopathy. *N Engl J Med*. 2025;392:33-44.
- Ioannou A, Patel RK, Razvi Y, et al. Impact of earlier diagnosis in cardiac ATTR amyloidosis over the course of 20 years. *Circulation*. 2022;146:1657-1670.
- Garcia-Pavia P, Kristen AV, Drachman B, et al. Survival in a real-world cohort of patients with transthyretin amyloid cardiomyopathy treated with tafamidis: an analysis from the Transthyretin Amyloidosis Outcomes Survey (THAOS). *J Card Fail*. 2025;31:525-533.
- Elliott P, Drachman BM, Gottlieb SS, et al. Long-term survival with tafamidis in patients with transthyretin amyloid cardiomyopathy. *Circ Heart Fail*. 2022;15:e008193.
- Masri A, Bhattacharya P, Medoff B, et al. A multicenter study of contemporary long-term tafamidis outcomes in transthyretin amyloid cardiomyopathy. *JACC CardioOncol*. 2025;7:282-293.
- Porcari A, Cappelli F, Nitsche C, et al. SGLT2 inhibitor therapy in patients with transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol*. 2024;83:2411-2422.
- Ioannou A, Massa P, Patel RK, et al. Conventional heart failure therapy in cardiac ATTR amyloidosis. *Eur Heart J*. 2023;44:2893-2907.
- Sperry BW, Hanna M, Shah SJ, Jaber WA, Spertus JA. Spironolactone in patients with an echocardiographic hfpef phenotype suggestive of cardiac amyloidosis. *JACC Heart Fail*. 2021;9:795-802.
- Maurer MS, Hanna M, Grogan M, et al. Genotype and phenotype of transthyretin cardiac amyloidosis: THAOS (Transthyretin Amyloid Outcome Survey). *J Am Coll Cardiol*. 2016;68:161-172.
- Volpentesta E, Kharoubi M, Donadio C, et al. Phenotype and prognostic factors in geriatric and non-geriatric patients with transthyretin cardiomyopathy. *ESC Heart Fail*. 2024;11:3814-3832.
- Reixach N, Deechongkit S, Jiang X, Kelly JW, Buxbaum JN. Tissue damage in the amyloidoses: transthyretin monomers and nonnative oligomers are the major cytotoxic species in tissue culture. *Proc Natl Acad Sci U S A*. 2004;101:2817-2822.
- Griffin JM, Rosenblum H, Maurer MS. Pathophysiology and therapeutic approaches to cardiac amyloidosis. *Circ Res*. 2021;128:1554-1575.
- Maurer MS, Witteles RM, Garcia-Pavia P, et al. Impact of heart failure severity on vutrisiran efficacy in transthyretin amyloidosis with cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1927-1939.
- Gupta V, Yanek LR, Shankar B, et al. A comparison of single versus combination mechanism treatment for transthyretin amyloid cardiomyopathy. *Amyloid*. 2026;33:79-81. <https://doi.org/10.1080/13506129.2025.2570095>
- Gillmore JD, Damy T, Fontana M, et al. A new staging system for cardiac transthyretin amyloidosis. *Eur Heart J*. 2018;39:2799-2806.
- Grogan M, Scott CG, Kyle RA, et al. Natural history of wild-type transthyretin cardiac amyloidosis and risk stratification using a novel staging system. *J Am Coll Cardiol*. 2016;68:1014-1020.
- Cheng RK, Levy WC, Vasbinder A, et al. Diuretic dose and NYHA functional class are independent predictors of mortality in patients with transthyretin cardiac amyloidosis. *JACC CardioOncol*. 2020;2:414-424.
- Manral P, Reixach N. Amyloidogenic and non-amyloidogenic transthyretin variants interact differently with human cardiomyocytes: insights into early events of non-fibrillar tissue damage. *Biosci Rep*. 2015;35:e00172.
- Zhao L, Buxbaum JN, Reixach N. Age-related oxidative modifications of transthyretin modulate its amyloidogenicity. *Biochemistry*. 2013;52:1913-1926.
- Writing Committee, Kittleson MM, Ambardekar AV, et al. Transthyretin cardiac amyloidosis evaluation and management: 2025 ACC concise clinical guidance. *J Am Coll Cardiol*. 2026;87(5):549-565. <https://doi.org/10.1016/j.jacc.2025.09.004>
- Garcia-Pavia P, Bengel F, Brito D, et al. Expert consensus on the monitoring of transthyretin amyloid cardiomyopathy. *Eur J Heart Fail*. 2021;23:895-905.
- Ioannou A, Cappelli F, Emdin M, et al. Stratifying disease progression in patients with cardiac ATTR amyloidosis. *J Am Coll Cardiol*. 2024;83:1276-1291.
- Nies RJ, Ney S, Nies JF, et al. Outpatient diuretic intensification: a simple prognostic marker in cardiac transthyretin amyloidosis. *Clin Res Cardiol*. Published online March 4, 2025. <https://doi.org/10.1007/s00392-025-02617-4>.
- Zeldin L, Eichler JBS, Teruya SL, et al. Outpatient worsening of heart failure and mortality in transthyretin amyloid cardiomyopathy. *Eur J Heart Fail*. 2025;27(12):2961-2970. <https://doi.org/10.1002/ehfj.3540>
- Bampatsias D, Wardhere A, Zeldin L, et al. Cardiac disease monitoring measures in patients with transthyretin amyloid cardiomyopathy treated with tafamidis. *Heart*. 2025;111(14):651-656. <https://doi.org/10.1136/heartjnl-2024-324826>
- Fontana M, Maurer MS, Gillmore JD, et al. Outpatient worsening heart failure in patients with transthyretin amyloidosis with cardiomyopathy in the HELIOS-B trial. *J Am Coll Cardiol*. 2025;85:753-761.
- Arora S, Patil NS, Strassle PD, et al. Amyloidosis and 30-day outcomes among patients with heart failure. *JACC CardioOncol*. 2020;2:710-718.
- Miller AB, Januzzi JL, O'Neill BJ, et al. Causes of cardiovascular hospitalization and death in patients with transthyretin amyloid cardiomyopathy (from the Tafamidis in Transthyretin Cardiomyopathy Clinical Trial [ATTR-ACT]). *Am J Cardiol*. 2021;148:146-150.
- Judge DP, Alexander KM, Cappelli F, et al. Efficacy of acoramidis on all-cause mortality and cardiovascular hospitalization in transthyretin amyloid cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1003-1014.
- Maurer MS, Kale P, Fontana M, et al. Patisiran treatment in patients with transthyretin cardiac amyloidosis. *N Engl J Med*. 2023;389:1553-1565.
- Minervini A, Barge-Caballero G, López-López A, et al. Transthyretin serum levels and clinical outcomes in patients with transthyretin amyloid cardiomyopathy. *Eur J Intern Med*. 2025;142:106350.
- Maurer MS, Judge DP, Gillmore JD, et al. Early increase in serum transthyretin by acoramidis

- independently predicts improved survival in TTR amyloid cardiomyopathy. *J Am Coll Cardiol*. 2025;85:1911-1923.
36. Bois RMD, Weycker D, Albera C, et al. Six-minute-walk test in idiopathic pulmonary fibrosis: test validation and minimal clinically important difference. *Am J Respir Crit Care Med*. 2011;183:1231-1237.
37. Shah MR, Hasselblad V, Gheorghiade M, et al. Prognostic usefulness of the six-minute walk in patients with advanced congestive heart failure secondary to ischemic or nonischemic cardiomyopathy. *Am J Cardiol*. 2001;88:987-993.
38. Ioannou A, Fumagalli C, Razvi Y, et al. Prognostic value of a 6-minute walk test in patients with transthyretin cardiac amyloidosis. *J Am Coll Cardiol*. 2024;84:43-58.
39. Mancini DM, Eisen H, Kussmaul W, et al. Value of peak exercise oxygen consumption for optimal timing of cardiac transplantation in ambulatory patients with heart failure. *Circulation*. 1991;83:778-786.
40. Ponikowski P, Francis DP, Piepoli MF, et al. Enhanced ventilatory response to exercise in patients with chronic heart failure and preserved exercise tolerance: marker of abnormal cardiorespiratory reflex control and predictor of poor prognosis. *Circulation*. 2001;103:967-972.
41. Bhutani D, Pan S, Latif F, et al. Cardiopulmonary exercise testing in patients with cardiac amyloidosis. *Clin Lymphoma Myeloma Leuk*. 2021;21:545-548.
42. Patel RK, Bandera F, Venneri L, et al. Cardiopulmonary exercise testing in evaluating transthyretin amyloidosis. *JAMA Cardiol*. 2024;9:367.
43. Patel RK, Bandera F, Venneri L, et al. Cardiopulmonary exercise testing with exercise echocardiography to assess treatment response in transthyretin amyloid cardiomyopathy. *JACC Heart Fail*. 2025;13:857-859.
44. Weldring T, Smith SMS. Article commentary: patient-reported outcomes (PROs) and patient-reported outcome measures (PROMs). *Health Serv Insights*. 2013;6:HSI.S11093.
45. Hanna M, Damy T, Grogan M, et al. Impact of tafamidis on health-related quality of life in patients with transthyretin amyloid cardiomyopathy (from the Tafamidis in Transthyretin Cardiomyopathy Clinical Trial). *Am J Cardiol*. 2021;141:98-105.
46. Nativi-Nicolau J, Judge DP, Hoffman JE, et al. Natural history and progression of transthyretin amyloid cardiomyopathy: insights from ATTR-ACT. *ESC Heart Fail*. 2021;8:3875-3884.
47. Levy JC, Teruya SL, Mirabal A, Wats K, Maurer MS, Bampatsias D. Worsening Kansas City Cardiomyopathy Questionnaire overall summary score (KCCQ-OS) is independently associated with increased risk of adverse outcomes in transthyretin cardiac amyloidosis (ATTR-CA). *J Am Coll Cardiol*. 2025;85(suppl):1179.
48. Cyrille NB, Goldsmith J, Alvarez J, Maurer MS. Prevalence and prognostic significance of low QRS voltage among the three main types of cardiac amyloidosis. *Am J Cardiol*. 2021;4(114):1089-1093.
49. Cipriani A, De Michieli L, Porcari A, et al. Low QRS voltages in cardiac amyloidosis. *JACC CardioOncol*. 2022;4:458-470.
50. Sperry BW, Vranian MN, Hachamovitch R, et al. Are classic predictors of voltage valid in cardiac amyloidosis? A contemporary analysis of electrocardiographic findings. *Int J Cardiol*. 2016;214:477-481.
51. Argirò A, Zampieri M, Mazzoni C, et al. Progression and prognostic significance of electrocardiographic findings in patients with cardiac amyloidosis. *ESC Heart Fail*. 2025;12:809-818.
52. Sanchis K, Cariou E, Colombat M, et al. Atrial fibrillation and subtype of atrial fibrillation in cardiac amyloidosis: clinical and echocardiographic features, impact on mortality. *Amyloid*. 2019;26:128-138.
53. Donnellan E, Wazni OM, Hanna M, et al. Atrial fibrillation in transthyretin cardiac amyloidosis. *JACC Clin Electrophysiol*. 2020;6:1118-1127.
54. Witteles R, Jefferies JL, Kapa S, et al. Atrial fibrillation as a prognostic factor for all-cause mortality in patients with transthyretin amyloid cardiomyopathy. *JACC CardioOncol*. 2024;6:592-598.
55. Chacko L, Karia N, Venneri L, et al. Progression of echocardiographic parameters and prognosis in transthyretin cardiac amyloidosis. *Eur J Heart Fail*. 2022;24:1700-1712.
56. Antonelli J, Neveu A, Kosmala W, et al. Evolution and prognostic value of left ventricular deformation and myocardial work parameters in transthyretin amyloid cardiomyopathy. *Eur Heart J Cardiovasc Imaging*. 2024;25:469-479.
57. Rettl R, Duca F, Binder C, et al. Impact of tafamidis on myocardial strain in transthyretin amyloid cardiomyopathy. *Amyloid*. 2023;30:127-137.
58. Giblin GT, Cuddy SAM, González-López E, et al. Effect of tafamidis on global longitudinal strain and myocardial work in transthyretin cardiac amyloidosis. *Eur Heart J Cardiovasc Imaging*. 2022;23:1029-1039.
59. Rettl R, Mann C, Duca F, et al. Tafamidis treatment delays structural and functional changes of the left ventricle in patients with transthyretin amyloid cardiomyopathy. *Eur Heart J Cardiovasc Imaging*. 2022;23:767-780.
60. Jering KS, Fontana M, Lairez O, et al. Effects of vutrisiran on cardiac structure and function in patients with transthyretin amyloidosis with cardiomyopathy: secondary outcomes of the HELIOS-B trial. *Nat Med*. 2025;31:3560-3568.
61. Fontana M, García-Pavía P, Grogan M, et al. Coramitug, a humanized monoclonal antibody for the treatment of transthyretin amyloid cardiomyopathy: a phase 2, randomized, multicenter, double-blind, placebo-controlled trial. *Circulation*. 2026;153:214-225.
62. Martinez-Naharro A, Treibel TA, Abdel-Gadir A, et al. Magnetic resonance in transthyretin cardiac amyloidosis. *J Am Coll Cardiol*. 2017;70:466-477.
63. Dobner S, Bernhard B, Nink L, et al. Impact of tafamidis on myocardial function and CMR tissue characteristics in transthyretin amyloid cardiomyopathy. *ESC Heart Fail*. 2024;11:2759-2768.
64. Fontana M, Martinez-Naharro A, Chacko L, et al. Reduction in CMR derived extracellular volume with patisiran indicates cardiac amyloid regression. *JACC Cardiovasc Imaging*. 2021;14:189-199.
65. Writing Committee, Kittleson MM, Ruberg FL, et al. 2023 ACC expert consensus decision pathway on comprehensive multidisciplinary care for the patient with cardiac amyloidosis: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2023;81:1076-1126.
66. Adams D, Tournev IL, Taylor MS, et al. Efficacy and safety of vutrisiran for patients with hereditary transthyretin-mediated amyloidosis with polyneuropathy: a randomized clinical trial. *Amyloid*. 2023;30:1-9.
67. Benson MD, Waddington-Cruz M, Berk JL, et al. Inotersen treatment for patients with hereditary transthyretin amyloidosis. *N Engl J Med*. 2018;379:22-31.
68. Adams D, Gonzalez-Duarte A, O'Riordan WD, et al. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *N Engl J Med*. 2018;379:11-21.
69. Coelho T, Marques W Jr, Dasgupta NR, et al. Eplontersen for hereditary transthyretin amyloidosis with polyneuropathy. *JAMA*. 2023;330:1448-1458.
70. Waddington Cruz M, Amass L, Keohane D, Schwartz J, Li H, Gundapaneni B. Early intervention with tafamidis provides long-term (5.5-year) delay of neurologic progression in transthyretin hereditary amyloid polyneuropathy. *Amyloid*. 2016;23:178-183.
71. Keohane D, Schwartz J, Gundapaneni B, Stewart M, Amass L. Tafamidis delays disease progression in patients with early stage transthyretin familial amyloid polyneuropathy: additional supportive analyses from the pivotal trial. *Amyloid*. 2017;24:30-36.
72. Karam C, Mauermann ML, Gonzalez-Duarte A, et al. Diagnosis and treatment of hereditary transthyretin amyloidosis with polyneuropathy in the United States: recommendations from a panel of experts. *Muscle Nerve*. 2024;69:273-287.
73. Folkvaljon F, Gertz M, Gillmore JD, et al. Estimating meaningful differences in measures of neuropathic impairment, health-related quality of life, and nutritional status in patients with hereditary transthyretin amyloidosis. *Muscle Nerve*. 2025;71:96-107.
74. Louwsma J, Brunger AF, Bijzet J, et al. Neurofilament light chain, a biomarker for polyneuropathy in systemic amyloidosis. *Amyloid*. 2021;28:50-55.
75. Berends M, Brunger AF, Bijzet J, et al. Longitudinal analysis of serum neurofilament light chain levels as marker for neuronal damage in hereditary transthyretin amyloidosis. *Amyloid*. 2024;31:132-141.
76. Galosi E, Costanzo R, Forcina F, et al. Serum neurofilament light chain levels correlate with small fiber related parameters in patients with

hereditary transthyretin amyloidosis with polyneuropathy (ATTRv-PN). *Neural Sci.* 2024;45:5023-5032.

77. Lobato L, Rocha A. Transthyretin amyloidosis and the kidney. *Clin J Am Soc Nephrol.* 2012;7:1337-1346.

78. Ioannou A, Rauf MU, Patel RK, et al. Albuminuria in transthyretin cardiac amyloidosis: prevalence, progression and prognostic importance. *Eur J Heart Fail.* 2024;26:65-73.

79. Ioannou A, Razvi Y, Porcari A, et al. Kidney outcomes in transthyretin amyloid cardiomyopathy. *JAMA Cardiol.* 2025;10:50-58.

80. Sperry BW, Sultan MB, Gundapaneni B, Tai SS, Witteles RM. Effect of tafamidis on renal function

in patients with transthyretin amyloid cardiomyopathy in ATTR-ACT. *JACC CardioOncol.* 2024;6:300-306.

81. Ioannou A, Nitsche C, Porcari A, et al. Multi-organ dysfunction and associated prognosis in transthyretin cardiac amyloidosis. *J Am Heart Assoc.* 2024;13:e033094.

82. Aimo A, Castiglione V, Tomasoni D, et al. Gamma-glutamyltransferase independently predicts mortality and heart failure hospitalization in cardiac transthyretin amyloidosis. *Eur J Intern Med.* 2025;140:106380.

83. Sattianayagam P, Hawkins P, Gillmore J. Amyloid and the GI tract. *Expert Rev Gastroenterol Hepatol.* 2009;3:615-630.

84. Suhr O, Danielsson A, Holmgren G, Steen L. Malnutrition and gastrointestinal dysfunction as prognostic factors for survival in familial amyloidotic polyneuropathy. *J Intern Med.* 1994;235:479-485.

85. Yamamoto S, Wilczek HE, Nowak G, et al. Liver transplantation for familial amyloidotic polyneuropathy (FAP): a single-center experience over 16 years. *Am J Transplant.* 2007;7:2597-2604.

KEY WORDS amyloidosis, disease progression, heart failure, transthyretin amyloidosis